CROP RECOMMENDATION USING MACHINE LEARNING

Beginner Machine Learning Project Report

Algorithm	Decision Tree Classifier
Dataset size	2,200 records and 8 columns
Input features	7 soil and environmental measurements
Test accuracy	97.95% on the selected 20% test split

Prepared as a learning and portfolio project. The results reported here come from the model run in the project notebook.

1. Abstract

This project builds a supervised machine learning classifier that predicts a crop category from soil nutrient values and environmental measurements. The model uses nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, soil pH, and rainfall as input features. A Decision Tree Classifier was trained and evaluated using a train-test split. The observed test accuracy was 97.95%. This result applies to the selected dataset and test split; it does not establish performance under all real-world farming conditions.

2. Problem Statement

Choosing a crop involves considering soil nutrients and local environmental conditions. This project explores whether a machine learning model can learn patterns from a labelled dataset and suggest one of the crop categories represented in that dataset.

3. Objectives

• Load and inspect a crop recommendation dataset.
• Check dataset dimensions, data types, and missing values.
• Separate input features (X) from the target label (y).
• Split the data into training and testing sets.
• Train a Decision Tree Classifier and evaluate its predictions.
• Demonstrate a prediction using a sample input.

4. Dataset Description

The dataset loaded in the notebook contained 2,200 rows and 8 columns. It represented 22 crop labels, with 100 records for each label. The notebook check found no missing values in the eight columns.

Column	Meaning	Role
N	Nitrogen value	Input feature
P	Phosphorus value	Input feature
K	Potassium value	Input feature
temperature	Temperature	Input feature
humidity	Humidity	Input feature
ph	Soil pH	Input feature
rainfall	Rainfall	Input feature
label	Crop category	Target to predict

5. Methodology

Step 1 — Data inspection: The dataset was loaded into a pandas DataFrame. Its shape, columns, data types, unique crop labels, and missing values were checked.
Step 2 — Feature and target selection: The seven numeric columns were assigned to X, and the crop label column was assigned to y.
Step 3 — Train-test split: The data was split with 80% for training and 20% for testing, using random_state=42 and stratify=y. This produced 1,760 training records and 440 testing records.
Step 4 — Model training: A DecisionTreeClassifier with random_state=42 was fitted on X_train and y_train.
Step 5 — Evaluation: Predictions on X_test were compared with y_test using accuracy, a classification report,

and a confusion matrix.

6. Model and Evaluation Results

Algorithm: Decision Tree Classifier.
Training set: 1,760 records.
Test set: 440 records.
Observed test accuracy: 0.979545..., or 97.95%.
The classification report showed macro-average and weighted-average F1-scores rounded to 0.98. Some classes, including blackgram, lentil, and mothbeans, had lower recall or precision than several other classes. The confusion matrix visualised correct and incorrect predictions by crop category.

7. Sample Prediction

A sample input was passed to the trained model using the seven expected features. The model returned the prediction **rice**. This was a demonstration of how the trained classifier can be used; it should not be treated as a practical crop recommendation for a real farm without local agronomic validation.

8. Limitations

• The model was evaluated on one train-test split of one labelled dataset.
• High test accuracy on this dataset does not guarantee equivalent performance in different regions, seasons, soil types, or changing weather conditions.
• The dataset labels and measurements may not represent every factor that affects real crop choice.
• The project is an educational prototype, not a replacement for soil testing or advice from qualified local agricultural experts.

9. Future Improvements

• Compare the Decision Tree with other suitable classification algorithms.
• Use cross-validation to estimate performance more robustly.
• Investigate the classes with lower precision or recall.
• Validate the model on independent, region-specific agricultural data.
• If deployed as an application, include clear guidance that recommendations require local validation.

10. Conclusion

This project demonstrates an end-to-end beginner machine learning workflow: loading and inspecting data, selecting features and a target, splitting the data, training a classifier, evaluating results, and generating a sample prediction. The Decision Tree Classifier achieved 97.95% accuracy on the selected test split. Further validation would be required before using the model for real agricultural decisions.

Reproducibility note

The notebook used pandas and scikit-learn, including train_test_split, DecisionTreeClassifier, accuracy_score, classification_report, and confusion_matrix. The dataset was loaded from a public Hugging Face-hosted CSV file during the session. Before formal submission, record the exact dataset source/version and retain the notebook used to produce the results.